



WESTMINSTER ASTRONOMICAL SOCIETY, INC. (WASI)



Membership News



Volume 42 – Number 4

Fall 2026

Message from our Society's President, Brooke Parkhurst

What is astronomy to you?

Each WASI member's unique vision has the power to inspire someone. Each one of us can share the excitement we feel about seeing into the night sky, and the reward is getting to see another person catch fire with how amazing the universe is. And even if you don't feel up to doing outreach, how fabulous is it to be able to hang out with a whole bunch of other people who think the stars are just ... stellar???

WASI is a living organism and whether you participate a little bit or a lot, that participation matters! Your participation in WASI, regardless of how small or how large, brings life and energy to this group and makes us the vital, energetic, crazy bunch that we are!

Bring a little joy to the group - come to an event this month, there are lots to choose from - and let us see that excited star face in person!

See you soon

Brooke

Notes and Thoughts

WASI Second Vice President Jeff Kretsch will be organizing some telescope clinics to help people get the most out of their equipment. He could use some help. For example, you might be accomplished at using the SkyMeadeCelestron AstroSeekerOrigin; share your expertise with others who have the same gear. Contact Jeff at jjkretsch@verizon.net.

Do you have any interest in radio astronomy? We're considering adding a subgroup if enough members express interest. With radio astronomy you can make observations during the day and even in cloudy weather, though there are no pretty pictures involved. Contact the secretary (secretary@westminsterastro.org) for more information.

This newsletter goes out quarterly. Do you have thoughts, insight or useful information for the members? Please consider submitting an article. Our survey last year showed that members really want more learning resources, so your expertise could be valuable to many. Send your ideas to secretary@westminsterastro.org.

ALCON 2026 Highlights; Cincinnati, OH/(held in) Covington, KY (8/13-8/16/2026); Hosted by Cincinnati Astronomical Society

WASI members present: Al & Laurie Ansorge, Matt Orsie

Conference Schedule:



CONFERENCE SCHEDULE

All events are held in the Covington Ballroom unless otherwise noted.

All presentation sessions are set to end at 50 minutes after they begin to allow a 10 minute break between presenters.

Wednesday 8/12

Early Conference Registration Check-in, Registration Check-in & Hospitality Table, Outside Covington Ballroom at the Marriott RiverCenter.

Welcome Dinner and Social at Molly Malone's Irish Pub - Covington KY (***ticketed activity***)

Thursday 8/13

8:00	Registration Check-in & Hospitality Table Opens Outside Covington Ballroom
8:30	
9:00	
9:30	
10:00	Conference Soft Start and Meet & Greet Brunch (***ticketed activity***) Marriott Atrium (Second Floor)
10:30	
11:00	Soft Start Presentation - John Ventre: The Mystery of the Stolen Lens Marriott Atrium (Second Floor)
11:30	
12:00	12:00PM - Vendor tables open
12:30	
1:00	Official Conference Start Welcome from AL President Chuck Allen and CAS President Bryan Simpson / General Session - David Knighton - Urban Observing: Yes You Can!
1:30	
2:00	Dr Nick Abel: Teaching Scientific Ideas to Non-science People
2:30	
3:00	Break
3:30	Dusty Segretto - Micrometeorites; A History and How-to
4:00	
4:30	Registration Check-in Closes
5:00	Vendor Tables Close for the Day
5:30	
6:00	StarBQ at the Cincinnati Museum Center (***ticketed activity***)
6:30	
7:00	For guests not attending please see our webpage for alternate ideas on dinner and other activities.
7:30	
8:00	
8:30	
9:00	
9:30	
10:00	StarBQ Ends - End of Scheduled Activities for Thursday (9:45PM)

Friday 8/14

Registration Check-in & Hospitality Table Re-Opens Outside Covington Ballroom

Vendor Tables Reopen - Kentucky Room (9:00)

Matthew Garrod / Jeff Rodriguez / John Blasing Citizen Science at CAS: Explorant observations, radio astronomy, and Radio JOVE.

BREAK

Astronomical League Business Meeting

Dr Wes Ryle Citizen Science Photometry: How Amateurs Support Modern Astronomy

Send-Off Instructions for Discovery Day

Explore Cincinnati Activity The Cincinnati Observatory Center (***ticketed activity***)

Underground Brewery Tour (***ticketed activity***)

Explore the city with CAS Ambassadors through a list of additional events. See list of possible ad hoc events at the hospitality desk or check the Explore Day event listing at [ALCON2026.org/ExploreDay](https://alcon2026.org/ExploreDay)



Vendors close for the day (7:00PM)

Featured Speaker - Pravera Hyson Water and Organics on Asteroids: Insights from Planetary Astronomy

Featured Speaker - Dr Shane Larson The Women of the Harvard Computers and the Birth of Modern Astronomy

ICE CREAM NIGHTCAP: COMPLIMENTARY ICE CREAM SUNDAY BAR - ALL WELCOME

End of Scheduled Activities for Friday (10:00 PM)

Saturday 8/15

Registration Check-in & Hospitality Table Re-Opens Outside Covington Ballroom

Vendor Tables ReOpen Kentucky Room (9:00AM)

Terry Endres / Dr Jeremy Huber / Emmy Bursk / Catherine Heile / Dr Nick Abel Panel on Engaging Young Scientists

BREAK

Jericho Kuehl: Smart Citizen Science 101: The Future of Amateur Astronomy

Bryan Simpson and Dieter Moeller: Bringing Pro Level Astronomy to Amateurs Through the Moeller Observatory / CAS Partnership

Isabel Tilton: CAS Outreach Programs Lunch Break - No formal offering. Guests discretion

1:00

Scott Harrington: Much Ado About Nebulae

2:00

Special Youth Award Sessions NYAA Winner Elizabeth Zhang: Discovery of Hidden Circumbinary Planets through Eclipse Timing

3:00

NYAA Runner-Up Michael Telesco A Solar-System-like Planetary Architecture around a Sun-like Star

4:00

Registration Check-In Table Closes Vendors Close

5:00

Social Hour in the 2nd floor atrium - GROUP PHOTO Complimentary Beer and Wine Bar. Music by Simply Strings. **DOOR PRIZE DRAWINGS** - (see page 23)

6:00

ALCON 2026 Keynote Banquet & Awards

7:00

(***ticketed activity***)

8:00

Keynote Speaker - Dr Rebecca Williams Snapshots of Mars Watery Past from NASA Rovers (8:00PM)

9:00

Astronomical League Awards

10:00

Conference Ends
** End of Scheduled Conference Activities **

Learned during conference that A.I. is used to crawl the web looking for copyright issues and folks are now suing for violations. Caution is shared with clubs to ensure that even speaker slides that might get reposted in videos don't pose a problem. A.L. had to do a small payout this year for a post done in 2014!

NOTICE TO ALL ATTENDEES OF ALCON 2026

DUE TO SEVERAL FACTORS, PHOTOGRAPHY AND VIDEO RECORDING OF ANY SPEAKER PRESENTATION IS NOT PERMITTED AT ANY TIME

PHOTOGRAPHY DURING AWARDS PRESENTATIONS IS PERMITTED

THE OFFICIAL PHOTOGRAPHERS MAY PROVIDE SOME PHOTOS OF OUR SPEAKERS FOR GUESTS AFTER THE CONCLUSION OF THE CONFERENCE

THANK YOU FOR FOR ADHERING TO THIS POLICY



All A.L. members who achieve "master observer" level are eligible to get a plaque, but must have it presented at an ALCON. There were 4 receiving theirs this year and Matt Orsie of WASI (from attaining the distinction ~25 years ago) was one. Pictured is Aaron Clevenson, Director of the Observing Programs Division





There was an imaging contest where entries were voted on by ALCON participants. The solar system category only had 2 entries.

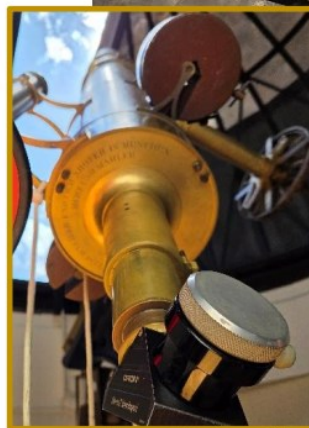
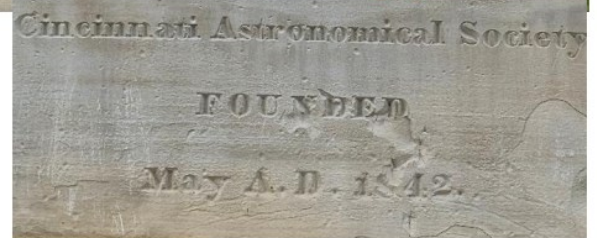


We toured the historic Cincinnati Observatory



4" Bausch & Lomb Refractor
Featuring an Equatorial Mount with a Mechanical Spring-Wound Clockwork
Circa 1910-1920
Donated by David Taylor, 2007. Restored by Bill Burton, 2009

Although this particular telescope was not used at the Cincinnati Observatory, it is very similar in age & design to the Alvan Clark CometSeeker Telescope that was used in this "cone observatory". That instrument was accidentally broken by a janitor and never replaced. The lens, which was cracked, survives and is in the Observatory's collection. The Comet Seeker's original mount is currently on display in the main building where it is supporting a larger Clark Telescope.



The new library telescope program equipment was on display.



The Star-B-Que was held at the Cincinnati Union Terminal with full access to the museum and Neil Armstrong collection.



At the various speaker-sessions I enjoyed learning about “micrometeorites,” (and how they’re all around us), the citizen science work members are doing at the Moeller Observatory 1.5 hours outside the city on exoplanets, the history of water in the solar system, women in the history of astronomy, and types of nebulae. The after party at the Moeller Observatory, featuring the 1-meter telescope was impressive too.



Expensive NB Filters Not Always Recommended

The narrowband filters we use in astrophotography do a great job of isolating specific wavelengths. Using, say, hydrogen alpha ($H\alpha$) can make an image really “pop.”

But be wary – with DSOs outside of the Milky Way some of the best NB filters may give you bogus data.

Distant objects are often red-shifted. The wavelength of light we receive is not what they radiate. The usual formula is this:

$$\frac{\Delta\lambda}{\lambda} = \frac{v}{c}$$

Where v is the object’s velocity moving away from us, c is the speed of light, λ is the nominal wavelength, and $\Delta\lambda$ is how much the wavelength has been red shifted.

Suppose you’re using very expensive Antila filters with a 3 nm passband. Any photon that’s more than 1.5 nm from the center of the passband will be blocked, or at least significantly attenuated.

For $H\alpha$ with a nominal wavelength of 656 nm, the surprising result is that the light from an object moving from us faster than 686 km/s will be blocked by that expensive filter.

Hubble showed us that, due to the expansion of the universe, an object’s distance in megaparsecs (1 Mpc is 3.26 light years) is:

$$d = \frac{v}{H_0}$$

Where H_0 is the Hubble constant, a number around 67 to 73 $\text{km s}^{-1} \text{Mpc}^{-1}$. For an object whose $H\alpha$ emissions will be blocked by that 3 nm filter, that corresponds to a distance of just 32 light years.

Holy crap! That means we can’t use an $H\alpha$ filter on, for example, M31! After all, Andromeda is 2.2 *million* light years away.

But M31 is gravitationally bound to us and is approaching at around 300 km/s, so this analysis doesn’t apply. That’s true for DSOs in the local group like M33 and the Clouds of Magellan.

But a lot of popular targets are receding rather rapidly due to the universe’s expansion. I surfed around Astrobin and found a lot of distant objects imaged with NB filters that shouldn’t be, as the filters were blocking the $H\alpha$, OIII and SII wavelengths. Those filters were picking up something – probably just the red signal for $H\alpha$ and SII, for example. In processing the astrophotographer was assigning a meaning to the data that wasn’t scientifically sound.

I had ChatGPT look for some of the most popular distant targets where filters may pervert the data, using SIMBAD to get observed redshifts to compute whether using narrowband filters made sense. Here’s the data for three filter passbands:

Popularity proxy rank	Galaxy	Observed redshift z	cz (km/s)	Observed H α (nm)	H α shift (nm)	3 nm centered filter	5 nm centered filter	7 nm centered filter
1	M100 / NGC 4321	0.005240	1571	659.72	3.44	OUTSIDE nominal FWHM	OUTSIDE nominal FWHM	Inside nominal FWHM
2	NGC 7331	0.002722	816	658.07	1.79	OUTSIDE nominal FWHM	Inside nominal FWHM	Inside nominal FWHM
3	NGC 4565	0.004165	1249	659.01	2.73	OUTSIDE nominal FWHM	OUTSIDE nominal FWHM	Inside nominal FWHM
4	M104 / NGC 4594	0.003416	1024	658.52	2.24	OUTSIDE nominal FWHM	Inside nominal FWHM	Inside nominal FWHM
5	NGC 3628	0.002880	863	658.17	1.89	OUTSIDE nominal FWHM	Inside nominal FWHM	Inside nominal FWHM
6	M87 / NGC 4486	0.004283	1284	659.09	2.81	OUTSIDE nominal FWHM	OUTSIDE nominal FWHM	Inside nominal FWHM
7	M109 / NGC 3992	0.003490	1046	658.57	2.29	OUTSIDE nominal FWHM	Inside nominal FWHM	Inside nominal FWHM
8	NGC 4725	0.004023	1206	658.92	2.64	OUTSIDE nominal FWHM	OUTSIDE nominal FWHM	Inside nominal FWHM
9	NGC 3521	0.002672	801	658.03	1.75	OUTSIDE nominal FWHM	Inside nominal FWHM	Inside nominal FWHM
10	M95 / NGC 3351	0.002595	778	657.98	1.70	OUTSIDE nominal FWHM	Inside nominal FWHM	Inside nominal FWHM
11	M96 / NGC 3368	0.002992	897	658.24	1.96	OUTSIDE nominal FWHM	Inside nominal FWHM	Inside nominal FWHM
12	M65 / NGC 3623	0.002692	807	658.05	1.77	OUTSIDE nominal FWHM	Inside nominal FWHM	Inside nominal FWHM
13	NGC 1365	0.005457	1636	659.86	3.58	OUTSIDE nominal FWHM	OUTSIDE nominal FWHM	OUTSIDE nominal FWHM
14	NGC 1300	0.005260	1577	659.73	3.45	OUTSIDE nominal FWHM	OUTSIDE nominal FWHM	Inside nominal FWHM
15	NGC 7479	0.007942	2381	661.49	5.21	OUTSIDE nominal FWHM	OUTSIDE nominal FWHM	OUTSIDE nominal FWHM
16	NGC 7814	0.003513	1053	658.59	2.31	OUTSIDE nominal FWHM	Inside nominal FWHM	Inside nominal FWHM
17	NGC 4567	0.007500	2248	661.20	4.92	OUTSIDE nominal FWHM	OUTSIDE nominal FWHM	OUTSIDE nominal FWHM
18	NGC 4568	0.007500	2248	661.20	4.92	OUTSIDE nominal FWHM	OUTSIDE nominal FWHM	OUTSIDE nominal FWHM
19	NGC 4559	0.002718	815	658.06	1.78	OUTSIDE nominal FWHM	Inside nominal FWHM	Inside nominal FWHM
20	NGC 772	0.008246	2472	661.69	5.41	OUTSIDE nominal FWHM	OUTSIDE nominal FWHM	OUTSIDE nominal FWHM
21	NGC 660	0.002850	854	658.15	1.87	OUTSIDE nominal FWHM	Inside nominal FWHM	Inside nominal FWHM
22	NGC 2146	0.002979	893	658.24	1.96	OUTSIDE nominal FWHM	Inside nominal FWHM	Inside nominal FWHM
23	NGC 3079	0.003689	1106	658.70	2.42	OUTSIDE nominal FWHM	Inside nominal FWHM	Inside nominal FWHM
24	NGC 4535	0.006550	1964	660.58	4.30	OUTSIDE nominal FWHM	OUTSIDE nominal FWHM	OUTSIDE nominal FWHM
25	M99 / NGC 4254	0.008030	2407	661.55	5.27	OUTSIDE nominal FWHM	OUTSIDE nominal FWHM	OUTSIDE nominal FWHM
26	M61 / NGC 4303	0.005224	1566	659.71	3.43	OUTSIDE nominal FWHM	OUTSIDE nominal FWHM	Inside nominal FWHM
27	M88 / NGC 4501	0.007609	2281	661.27	4.99	OUTSIDE nominal FWHM	OUTSIDE nominal FWHM	OUTSIDE nominal FWHM
28	NGC 5248	0.003838	1151	658.80	2.52	OUTSIDE nominal FWHM	OUTSIDE nominal FWHM	Inside nominal FWHM
29	NGC 5371	0.008530	2557	661.88	5.60	OUTSIDE nominal FWHM	OUTSIDE nominal FWHM	OUTSIDE nominal FWHM
30	NGC 5364	0.004146	1243	659.00	2.72	OUTSIDE nominal FWHM	OUTSIDE nominal FWHM	Inside nominal FWHM
31	NGC 5033	0.002919	875	658.20	1.92	OUTSIDE nominal FWHM	Inside nominal FWHM	Inside nominal FWHM
32	NGC 5005	0.003159	947	658.35	2.07	OUTSIDE nominal FWHM	Inside nominal FWHM	Inside nominal FWHM
33	NGC 2906	0.007138	2140	660.96	4.68	OUTSIDE nominal FWHM	OUTSIDE nominal FWHM	OUTSIDE nominal FWHM
34	NGC 2775	0.004597	1378	659.30	3.02	OUTSIDE nominal FWHM	OUTSIDE nominal FWHM	Inside nominal FWHM
35	NGC 3370	0.004270	1280	659.08	2.80	OUTSIDE nominal FWHM	OUTSIDE nominal FWHM	Inside nominal FWHM
36	NGC 3675	0.002557	767	657.96	1.68	OUTSIDE nominal FWHM	Inside nominal FWHM	Inside nominal FWHM
37	NGC 3893	0.003259	977	658.42	2.14	OUTSIDE nominal FWHM	Inside nominal FWHM	Inside nominal FWHM
38	NGC 3938	0.002699	809	658.05	1.77	OUTSIDE nominal FWHM	Inside nominal FWHM	Inside nominal FWHM
39	NGC 4038 (Antennae)	0.005593	1677	659.95	3.67	OUTSIDE nominal FWHM	OUTSIDE nominal FWHM	OUTSIDE nominal FWHM
40	NGC 4039 (Antennae)	0.005474	1641	659.87	3.59	OUTSIDE nominal FWHM	OUTSIDE nominal FWHM	OUTSIDE nominal FWHM
41	NGC 4088	0.002502	750	657.92	1.64	OUTSIDE nominal FWHM	Inside nominal FWHM	Inside nominal FWHM
42	NGC 4157	0.002568	770	657.97	1.69	OUTSIDE nominal FWHM	Inside nominal FWHM	Inside nominal FWHM
43	NGC 4217	0.003400	1019	658.51	2.23	OUTSIDE nominal FWHM	Inside nominal FWHM	Inside nominal FWHM
44	NGC 4651	0.002685	805	658.04	1.76	OUTSIDE nominal FWHM	Inside nominal FWHM	Inside nominal FWHM
45	NGC 4698	0.003340	1001	658.47	2.19	OUTSIDE nominal FWHM	Inside nominal FWHM	Inside nominal FWHM
46	NGC 4753	0.003910	1172	658.85	2.57	OUTSIDE nominal FWHM	OUTSIDE nominal FWHM	Inside nominal FWHM
47	NGC 5775	0.005607	1681	659.96	3.68	OUTSIDE nominal FWHM	OUTSIDE nominal FWHM	OUTSIDE nominal FWHM
48	M102 / NGC 5866	0.002520	755	657.93	1.65	OUTSIDE nominal FWHM	Inside nominal FWHM	Inside nominal FWHM
49	NGC 5985	0.008050	2413	661.56	5.28	OUTSIDE nominal FWHM	OUTSIDE nominal FWHM	OUTSIDE nominal FWHM
50	NGC 7217	0.003170	950	658.36	2.08	OUTSIDE nominal FWHM	Inside nominal FWHM	Inside nominal FWHM

The good news is that the nebulas we love to image are all in the Milky Way and their NB emissions fall easily into the best filter passbands.

Most extragalactic objects are not gravitationally bound to our galaxy. Be wary about using NB filters on them! Use [SIMBAD](#) or a similar resource to get the redshift (z).

You'll notice in this chart "cz (km/s)" which is the object's speed away from us (z times the speed of light). Substituting that into the first equation on the previous page, we get:

$$\Delta\lambda = z\lambda$$

So, multiply the redshift times the nominal wavelength (656 nm for H α) and you'll get how much the emission has shifted due to the expansion of the universe. If it falls outside your filter's bandwidth, stick to the usual LRGB filters.

Wolf-Rayet Stars

The Cygnus starfield is a glorious site, dense with Milky Way stars. While visual observations of the region are breathtaking, an astrophoto reveals so much more. The following photo shows two fascinating objects, WR134 and WR 135, both “Wolf-Rayet” stars. Charles Wolf and Georges Rayet discovered three unusual stars in 1867, which stood out due to their odd broad emission lines, which we now know are produced in extremely-dense, rapidly expanding stellar winds. WR134 and WR 135 in this image are two of the three stars Wolf and Rayet found.

These are some of the hottest of all stars, with surface temperatures of some exceeding 200,000 degrees K. Some of these have a bolometric luminosity (brightness across all wavelengths) millions of times that of the Sun, but, alas, most of that is in the UV, outside our visual perception range.



Massive stars lose most of their outer envelope due to radiation-driven stellar winds, and in some cases, a companion strips material away. WR stars commonly lose 10^{-6} to 10^{-5} solar masses per year, orders of magnitude more than the Sun does. The products of fusion get exposed; typically we see helium and nitrogen (called WN stars) and helium and carbon (WC). But the fun doesn't last; after a few hundred thousand years the fusion reactions at the core can't compete with the staggering gravitational forces, and it dies in a core-collapse supernova. Depending on the mass, the result is a black hole or neutron star. So, a Wolf-Rayet star is really just a stage of life for some very massive stars.

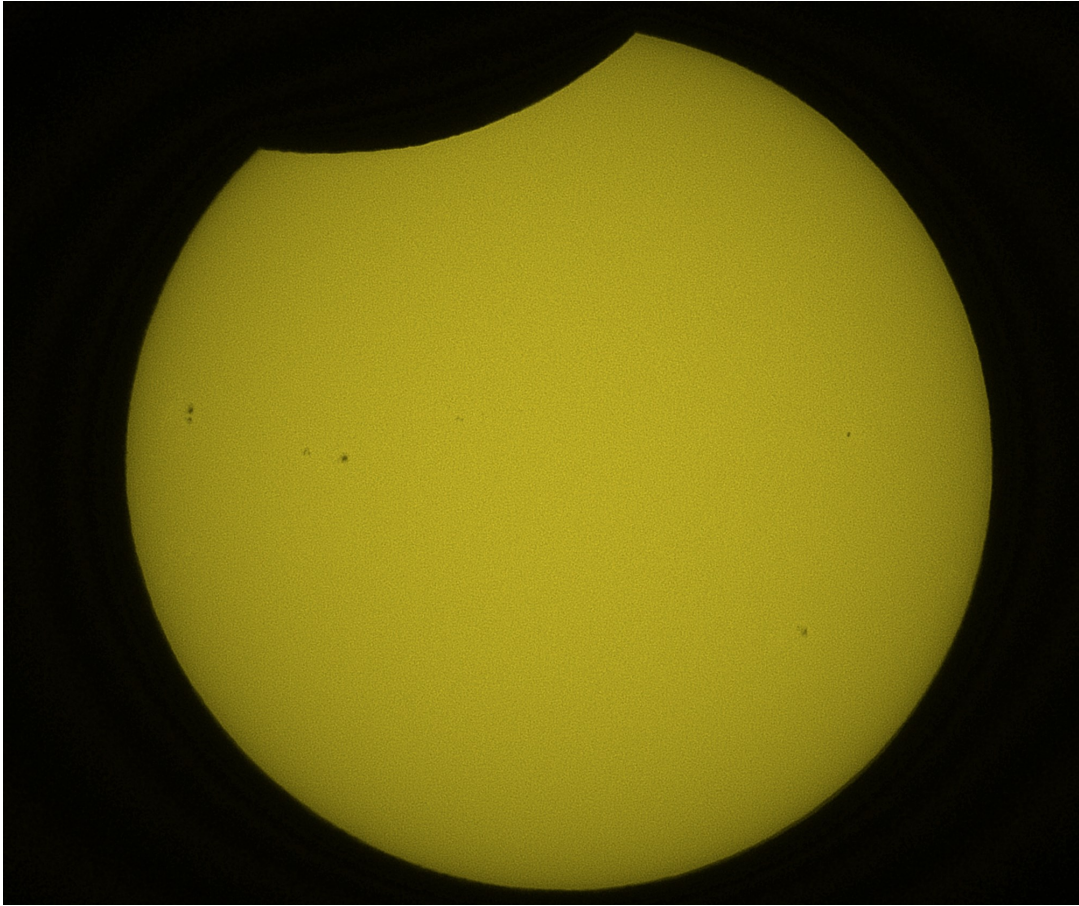
Astrophotos From Our Members

Matt Orsie sent this: Here's a 2.5 hr capture of my favorite northern hemisphere globular cluster, M5. Taken in early June. Seestar S50. Siril with 1.5x drizzle, and Photoshop.



Astrophotos From Our Members

Laurie Ansorge's Unistellar Odyssey Pro caught this nice, sharp image of the August eclipse:

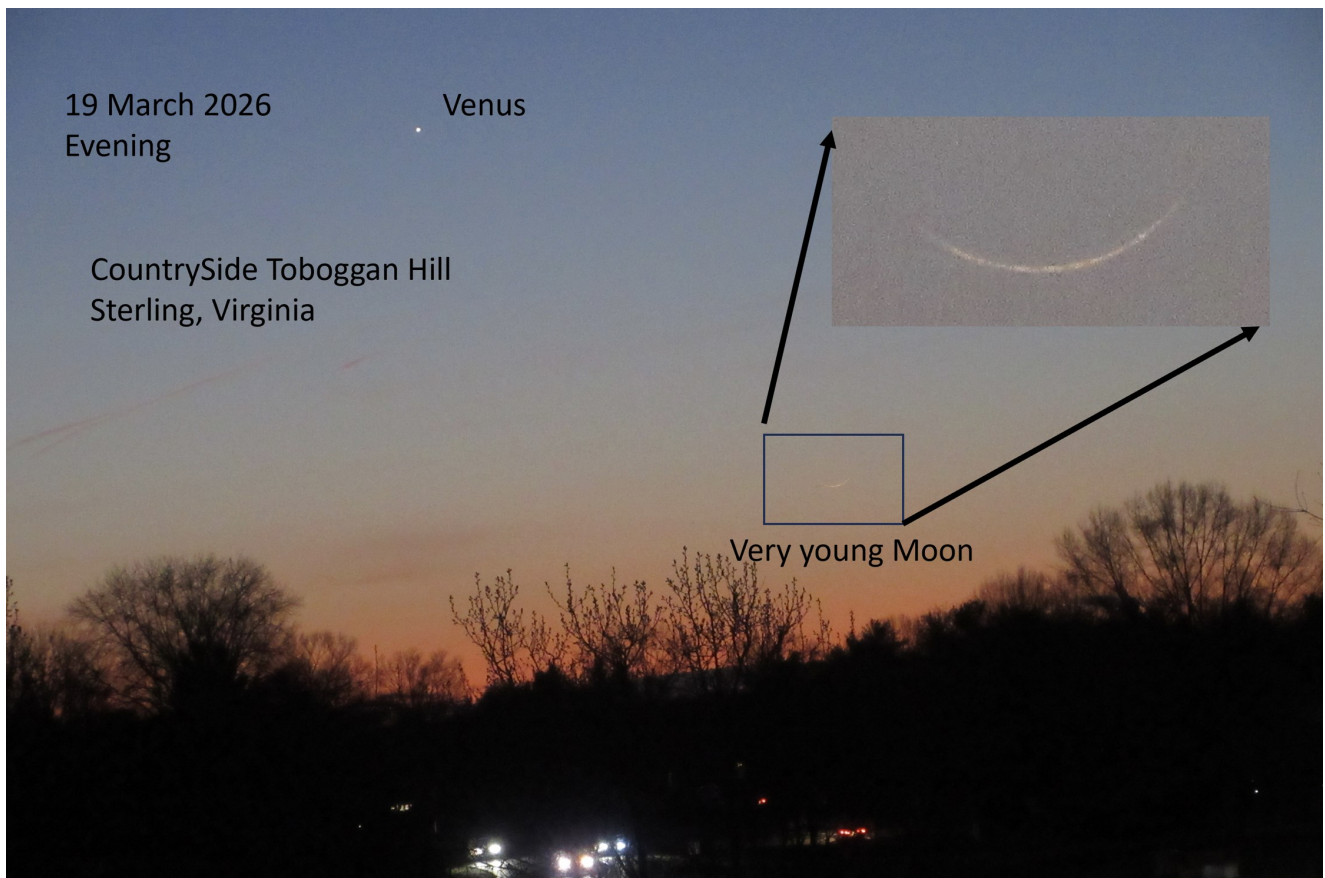


And her Celestron Origin with a new updated camera (678C) snagged the Whirlpool galaxy:

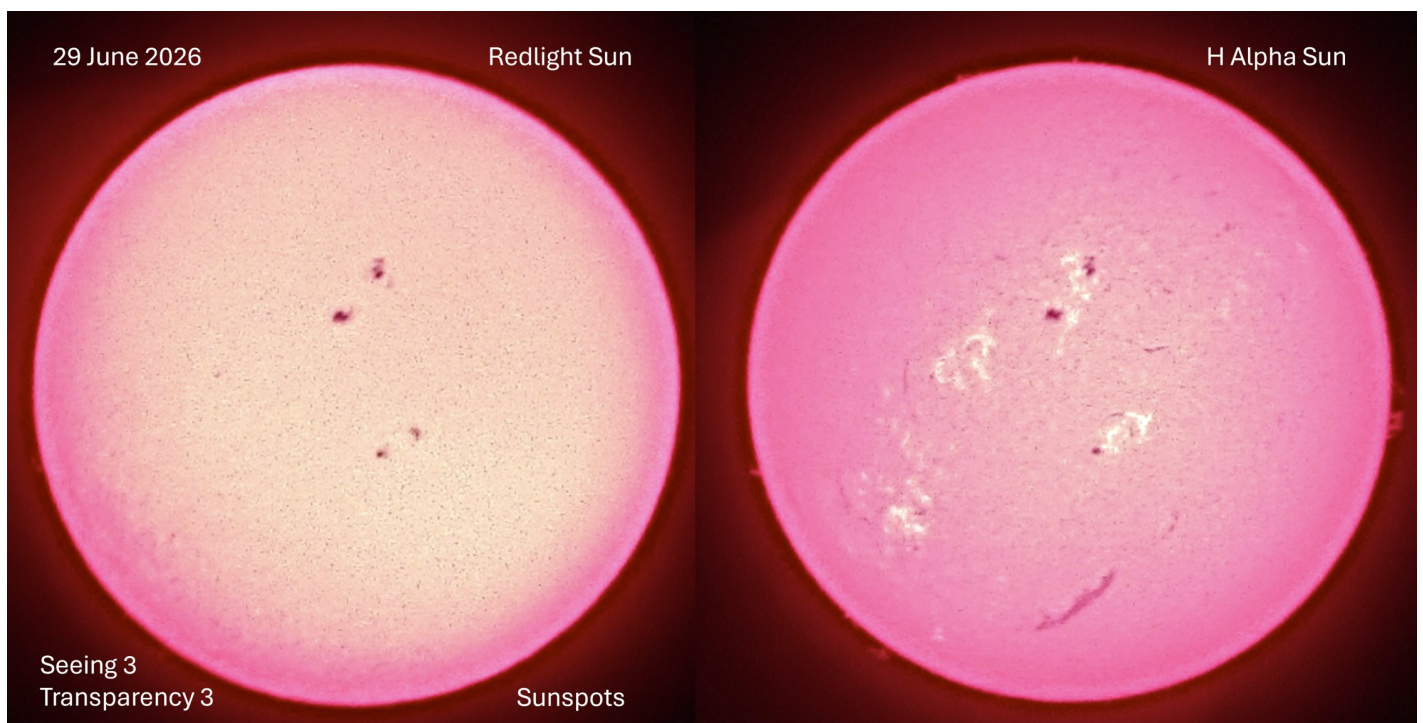


Astrophotos From Our Members

Jeff Kretsch got this amazing picture of a very, very young moon:



And he continues to dazzle us with both white light and hydrogen alpha solar images:



WASI FAQs

Library - Did you know we have over 700 books about astronomy in our WASI library? There are available to WASI members. Here's the complete card catalog: <https://westminsterastro.groups.io/g/main/files>.

Loaner telescopes - We also have a telescope lending library. If you'd like to borrow a scope, talk to Curt Roelle.

Astronomical League - All WASI members are also members of the Astronomical League. Check out their 80+ observing programs, many of which come with awards: <https://www.astroleague.org/>

Newsletter - Send pictures, articles, and ideas for the newsletter to secretary@westminsterastro.org.

Facebook - We're active and sharing images on our Facebook page, found here:



Join/Renew membership link: <https://www.westminsterastro.org/join-wasi/>

If you've already entered your contact information (renewing), skip the "database" link: <https://paypal.me/WAstroSInc>

Dues are payable via PayPal on the link above, by check or cash (and through your bank's on-line bill payment). Membership Dues are \$25/year for individuals or family, and youth under 18 is \$5/year. Or, save money and sign up for 2 years (\$40), or 4 years (\$80).

- On time payment means eligibility for the annual incentive.
- Access to the club's observatory (after training).
- Keep access to the members-only groups.io pages/information
- Receive members-only access/notifications on Night Sky Network
- Keep/get discount rates for popular astronomy magazines
- Borrow from the WASI scope/literature library

Files and club member correspondence & wiki links are found here: <https://westminsterastro.groups.io/g/main>. Remember to set your communication preferences.

Outreach/event calendar is found on: <https://nightsky.jpl.nasa.gov/index.cfm>. Set your communication preferences here as well.

Changed address, email or phone? Please update your information and send a message to the webmaster and/or treasurer@westminsterastro.org.

We meet monthly on the 2nd Wednesday of the month:
Back to Basics from 7:00 PM – 7:30PM; General Meeting 7:30PM – 9:30PM
Bear Branch Nature Center Carroll County; 300 John Owings Rd.; Westminster, MD 21158
Website: <https://www.westminsterastro.org/> (Zoom info for hybrid meetings)